ADITYA UNIVERSITY

Freshman Engineering Department
Department of Mathematics
Practice Questions

AY: 2025-26 **SEM-II**

Course Name & Code: Differential Equations and Vector Calculus (2501MA02).

Unit-IV: Vector Differentiation.

1. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$.
2. Find the angle between the surfaces $xy = z^2$ at (4, 1, 2) and (3, 3, -3).
3. Find the values of a and b such that the surfaces $ax^2 - byz = (a + 2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1, -1, 2)$.
4. Find the directional derivative of $f(x, y, z) = xy^3 + yz^3$ at $(2, -1, 1)$ in the direction of the vector $\bar{i} + 2\bar{j} + 2\bar{k}$.
5. Find the directional derivative of $\emptyset(x, y, z) = xy^2 + yz^3$ at $(2, -1, 1)$ in the direction of the normal to the surface $x\ log\ z - y^2 = -4$ at $(-1, 2, 1)$.
6. Find the directional derivative of $x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ where Q is the point $(5, 0, 4)$. Also find the magnitude of the maximum directional derivative.
7. Find the directional derivative of $xy^2 + yz^2 + zx^2$ along the tangent to the curve $x = t, y = t^2, z = t^3$ at the point $(1, 1, 1)$.
8. The temperature of points in a space is given by $T(x, y, z) = x^2 + y^2 - z$, a mosquito is located at (1,1, 2) desires to fly in such a direction that it will get warm as soon as possible. In what direction should it move. Also find what is the maximum rate of change in the temperature.
9. Find $div\ \bar{f}$ and $cur\ \bar{f}$, given that $\bar{f} = grad\ (x^3 + y^3 + z^3 - 3xyz)$.
10. Find $div\ \bar{F}$ and $curl\ \bar{F}$ at $(1, -1, 1)$, given that $\bar{F} = xy^2\,\bar{\imath} + 2x^2yz\bar{\jmath} - 3yz^2\,\bar{k}$.
11. Prove that $\text{div}(r^n\bar{r}) = (n + 3)r^n$, hence show that $\frac{\bar{r}}{r^3}$ is solenoidal.
12. Show that the vector field $\bar{F} = 2xyz^2\bar{\imath} + (x^2z^2 + z\cos yz)\bar{\jmath} + (2x^2yz + y\cos yz)\,\bar{k}$ is irrotational and find the potential function.
13. Show that the vector field $\bar{F} = (x^2 - yz)\bar{\imath} + (y^2 - zx)\bar{\jmath} + (z^2 - xy)\bar{k}$ is irrotational and hence find the scalar potential function corresponding to it.
14. Find the constants a, b, c so that the following vector $\bar{A}$ is irrotational. Also find its scalar potential function. Given that $\bar{A} = (x + 2y + az)\bar{\imath} + (bx - 3y - z)\bar{\jmath} + (4x + cy + 2z)\bar{k}$.
15. Prove that (i) $div\big(curl\ \bar{f}\big) = 0$ (ii) $curl(grad\ \emptyset) = \bar{0}$.
16. Show that $div(grad\ r^n) = \nabla^2(r^n) = n(n + 1)r^{n-2}$ where $r = |\bar{r}|$.
17. Show that $\nabla^2[f(r)] = f''(r) + \frac{2}{r}f'(r)$ where $r = |\bar{r}|$.